TECHNICAL BENCHMARK INTELLIGENCE / 2026

AMD AI Factory Mission Control Benchmark Atlas

A consolidated visual report of recorded AMD Instinct MI300X inference tests, scheduler optimization, end-to-end request latency, and 16-job dispatch behavior.

REAL MI300X RUNS	JOB TRACE ROWS	PROFILER PATHS
4 documented	16 timed jobs	3 paths x 3 trials

Evidence boundary: This PDF is a preview constructed from benchmark summaries and job records provided earlier. It is not a re-analysis of the complete raw archive on the Windows project. Use the bundled offline generator to index every stored *v2_** run, latency profiler, Scheduler Lab experiment, reliability log, and monitoring record.

Test configuration: one AMD Instinct MI300X inference endpoint, Qwen/Qwen3-0.6B, maximum two concurrent controller jobs, vLLM on ROCm. The first four-job benchmark uses different throughput accounting than subsequent v2 runs.

Publication: All values are descriptive measurements from the provided test outputs. Single runs and sparse GPU polls are not sufficient to establish reproducibility or hardware saturation.

01 / Executive benchmark index

Run	Jobs	P95 queue (s)	P95 E2E (s)	Client tok/s	Failures
4 jobs initial v0.3	4	n/a	n/a	not comparable	0
8 jobs before v0.5	8	11.2829	12.7369	49.607	0
8 jobs after v0.5	8	4.6745	5.8264	69.661	0
16 jobs after v0.5	16	13.7397	15.3572	75.68	0

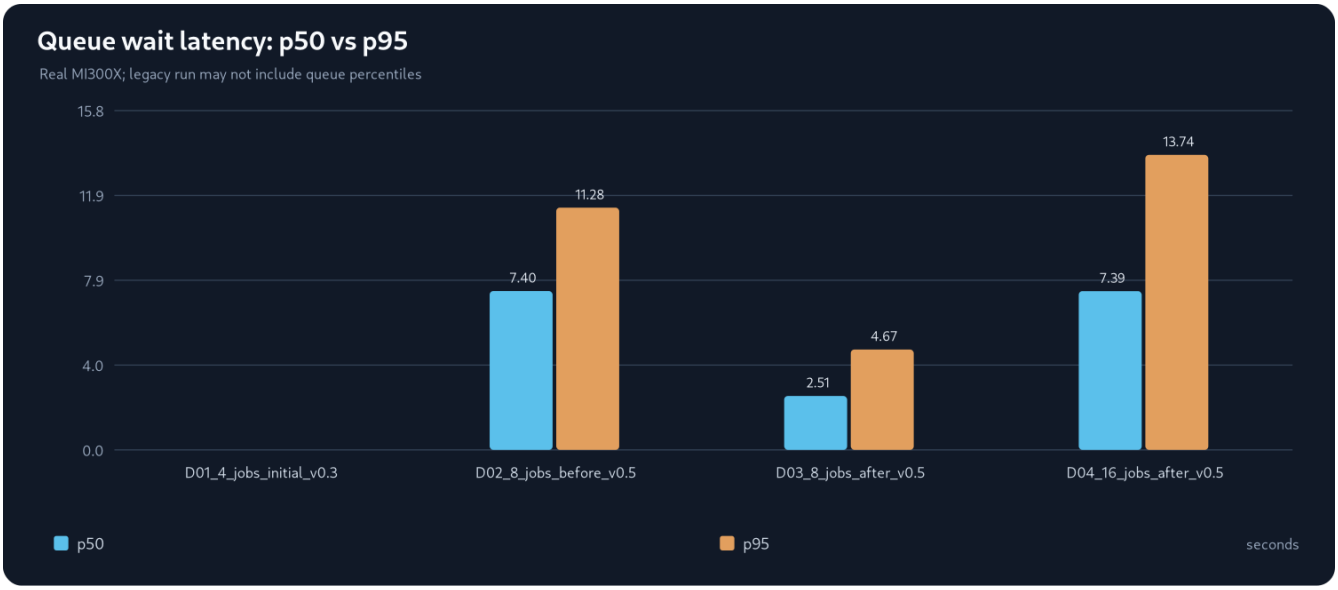
02 / Measured optimization

For an eight-request workload, the v0.5 scheduling change is associated with a **58.57% decrease in p95 queue waiting time**, a **54.26% decrease in p95 end-to-end latency**, and a **40.43% increase in client-observed output throughput**. The inference HTTP call duration was essentially unchanged, so the observed gains largely concern waiting and controller lifecycle overhead.

METRIC	BEFORE v0.5	AFTER v0.5	DESCRIPTIVE CHANGE
p95 queue wait	11.2829 s	4.6745 s	-58.57%
p95 end-to-end	12.7369 s	5.8264 s	-54.26%
p95 inference HTTP	0.2073 s	0.2110 s	+1.8% duration
Client output throughput	49.607 tok/s	69.661 tok/s	+40.43%

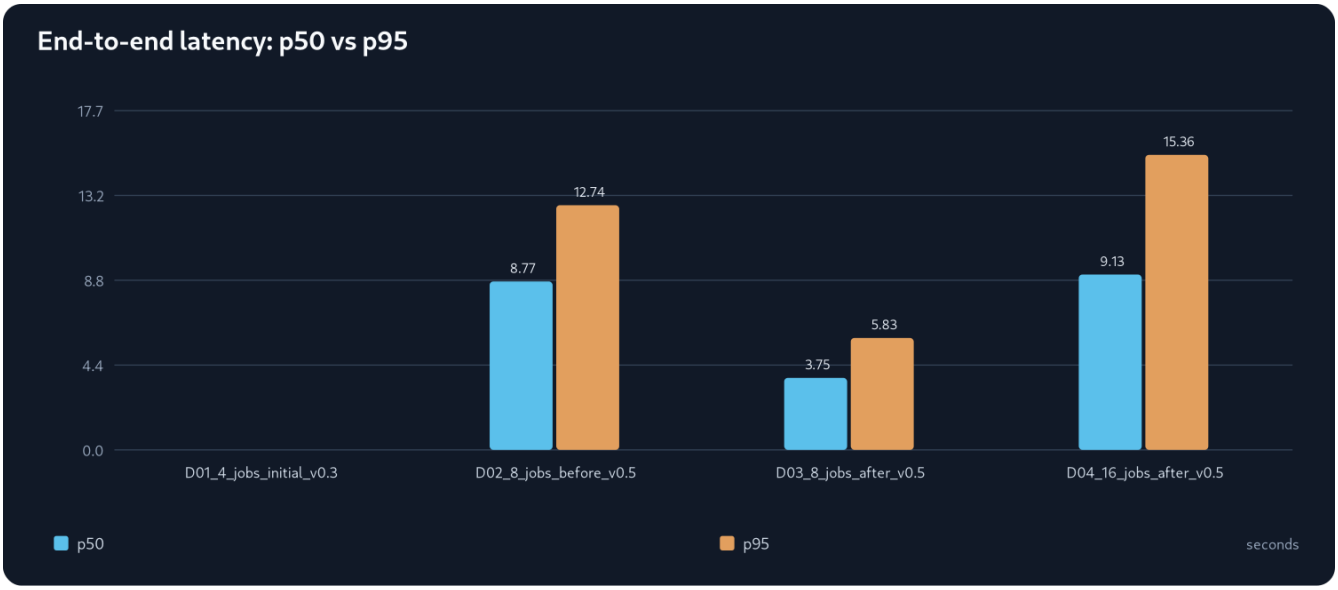
This is a one-run-before vs one-run-after comparison. It should not be presented as a confidence interval or statistical significance test.

03 / Queue waiting latency



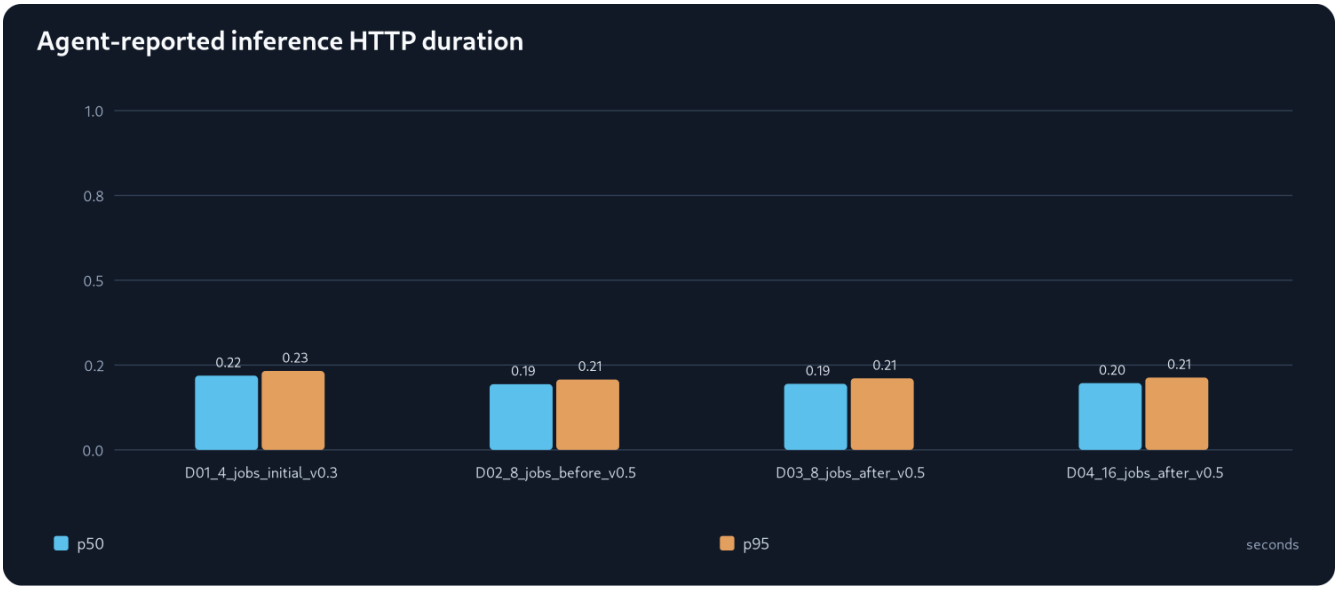
p50 and p95 waiting times, where available. Queue p95 rises at 16 requests.

04 / End-to-end latency



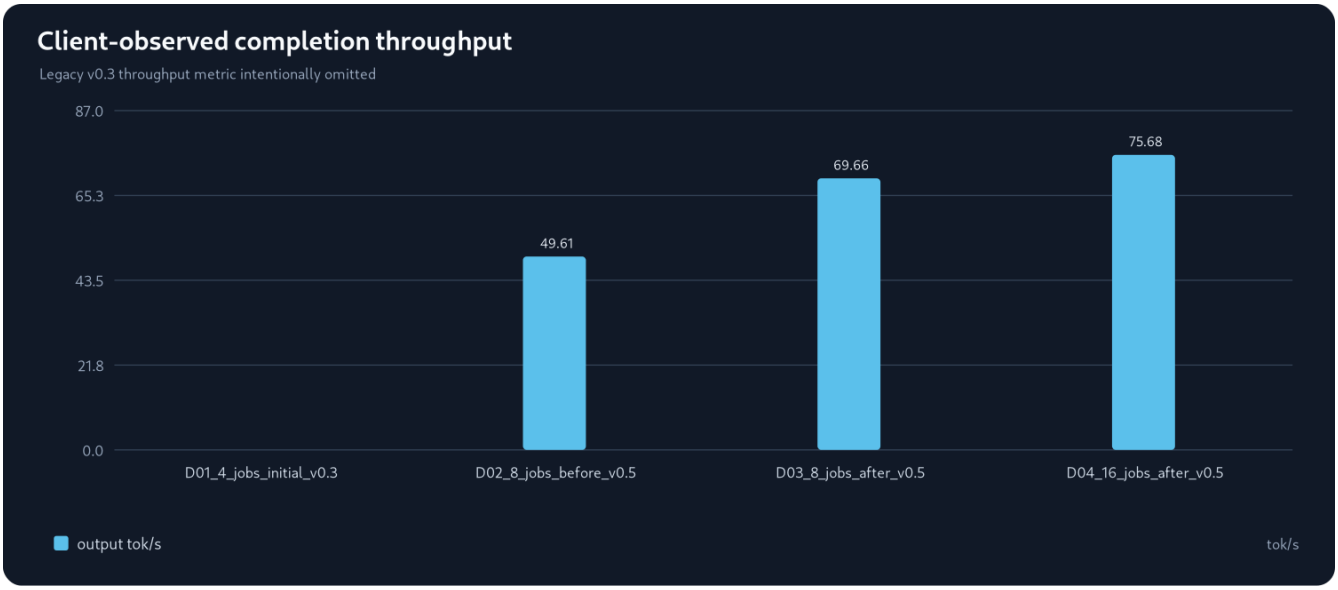
p50 and p95 controller lifecycle latency, in seconds.

05 / Inference HTTP duration



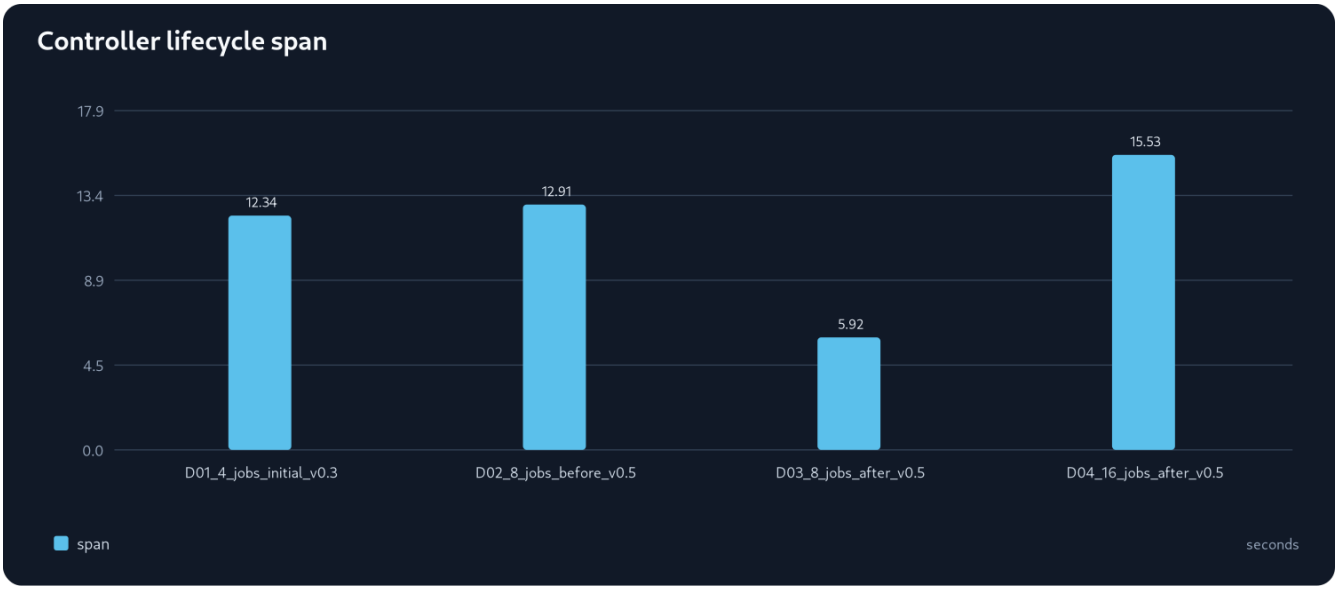
GPU-facing HTTP call duration stays near 0.2 seconds in the reported runs.

06 / Client throughput



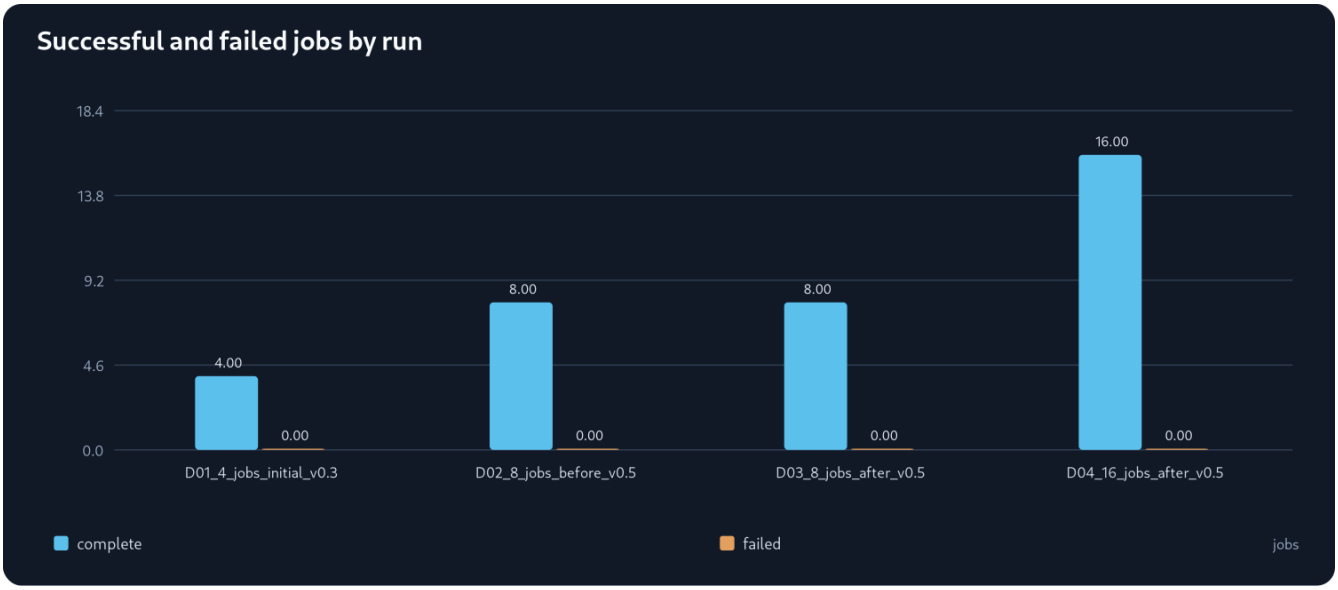
Output tokens per observed client wall second. Legacy v0.3 omitted because denominator differs.

07 / Controller lifecycle span



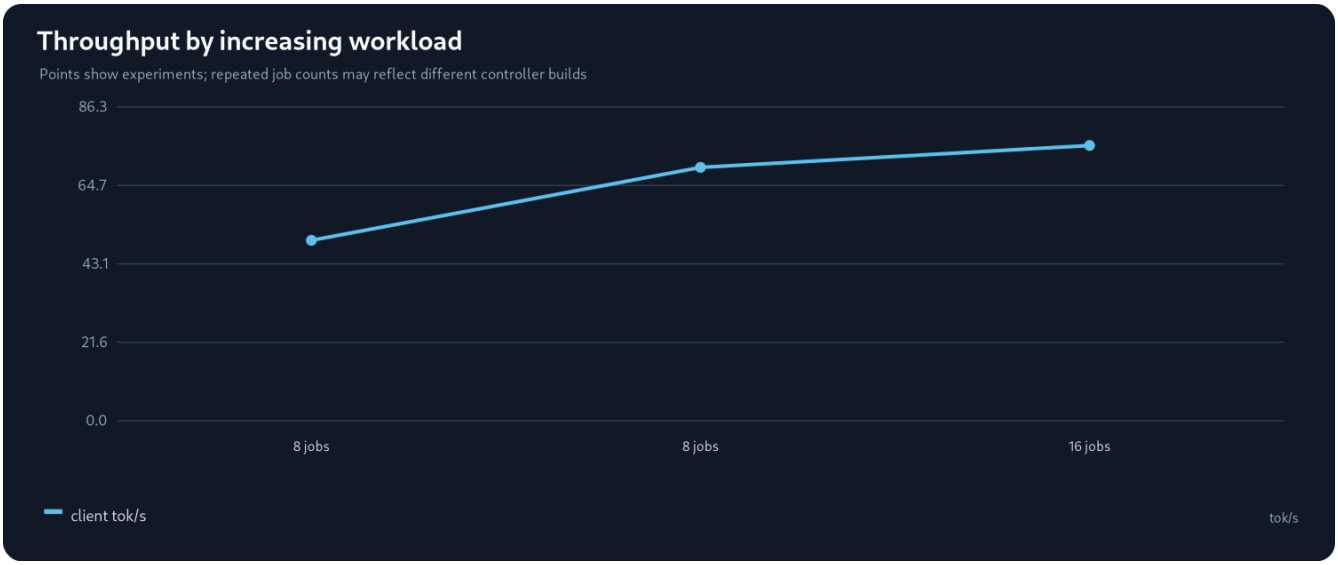
Total time between earliest create and latest completion; not isolated GPU time.

08 / Completion reliability



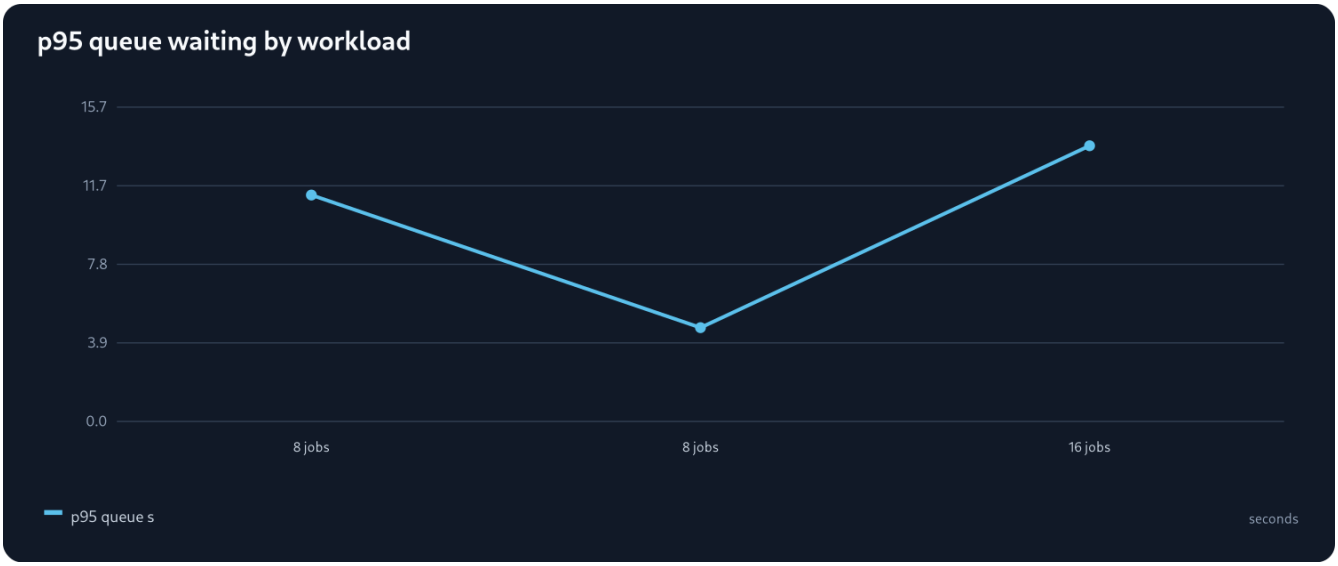
No reported failures in the selected MI300X benchmarks.

09 / Load vs throughput



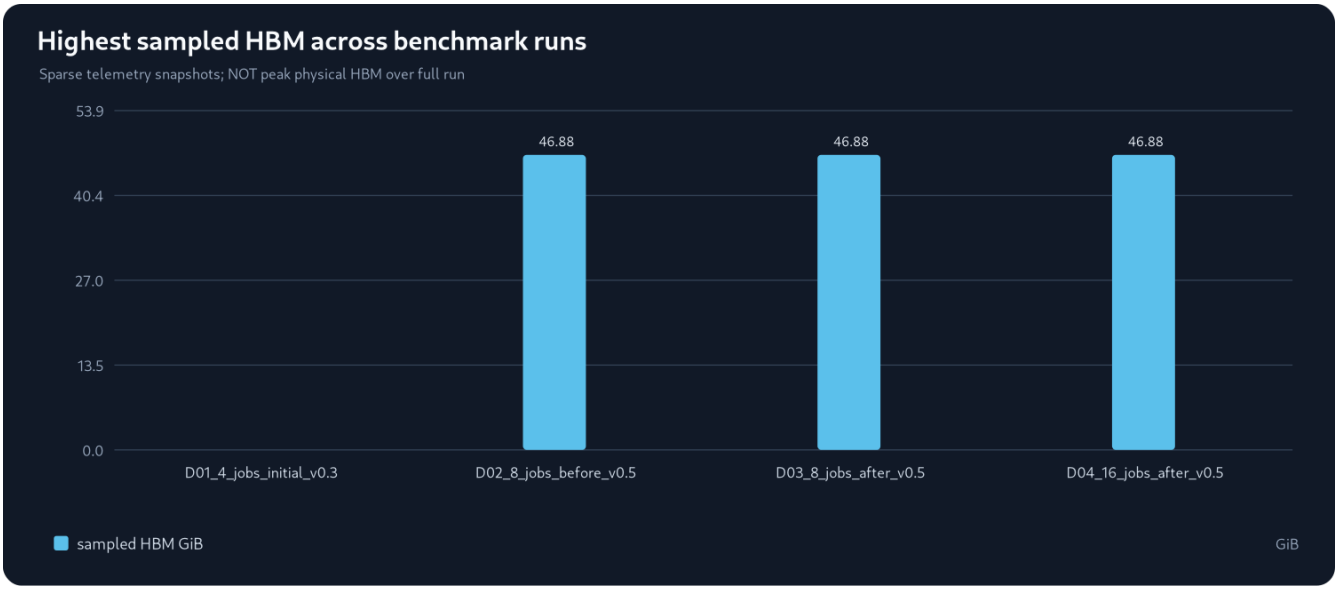
Descriptive scaling; optimized 8- and 16-job runs share controller version.

10 / Load vs queue pressure



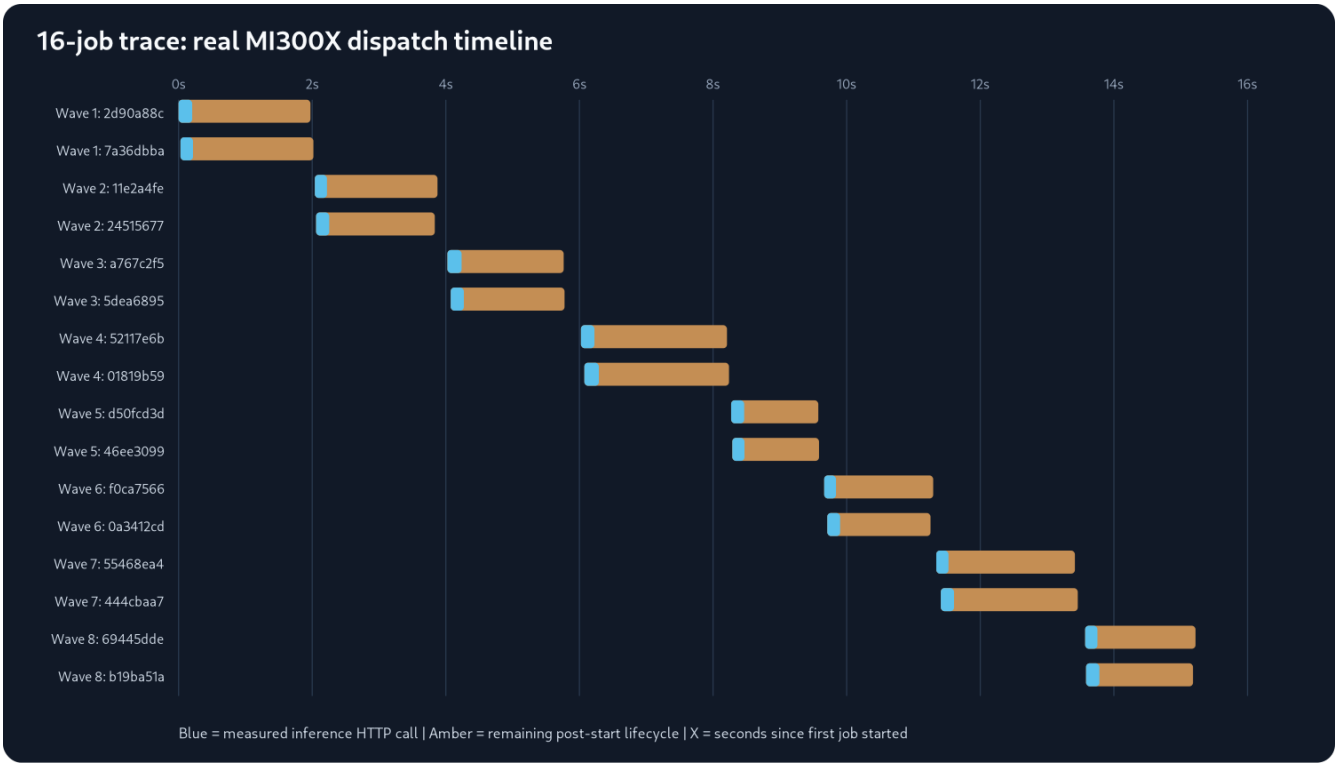
Illustrates rising wait time under higher request pressure.

11 / HBM telemetry snapshots



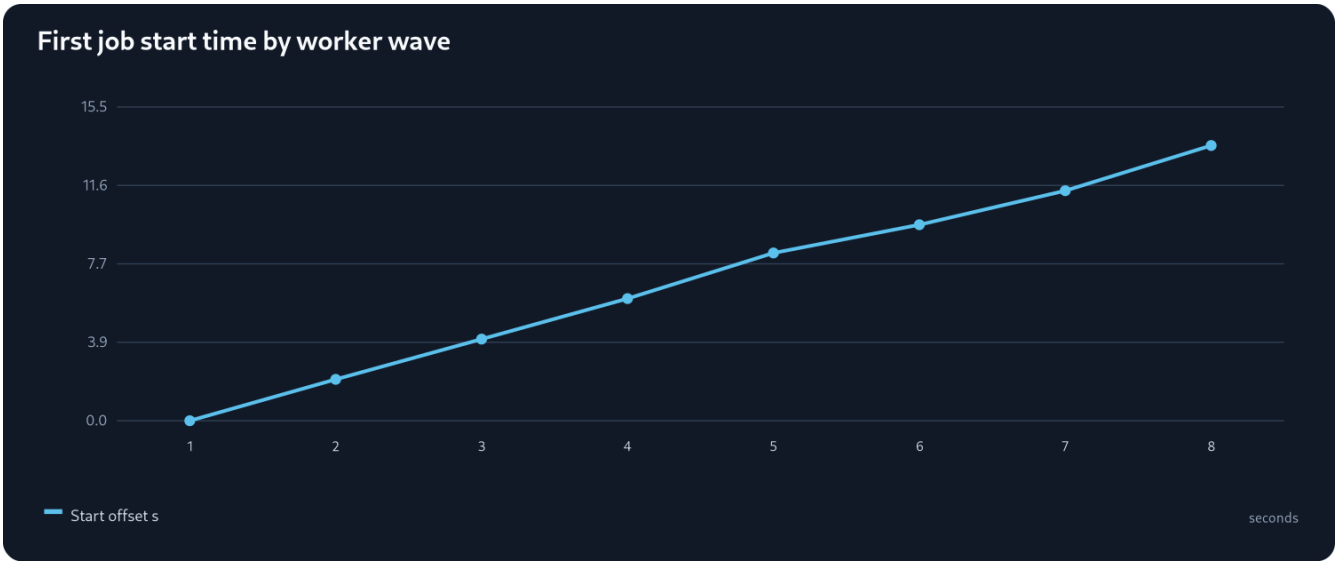
46.885 GiB is a sampled value, not necessarily the true hardware peak.

12 / Per-job Gantt timeline



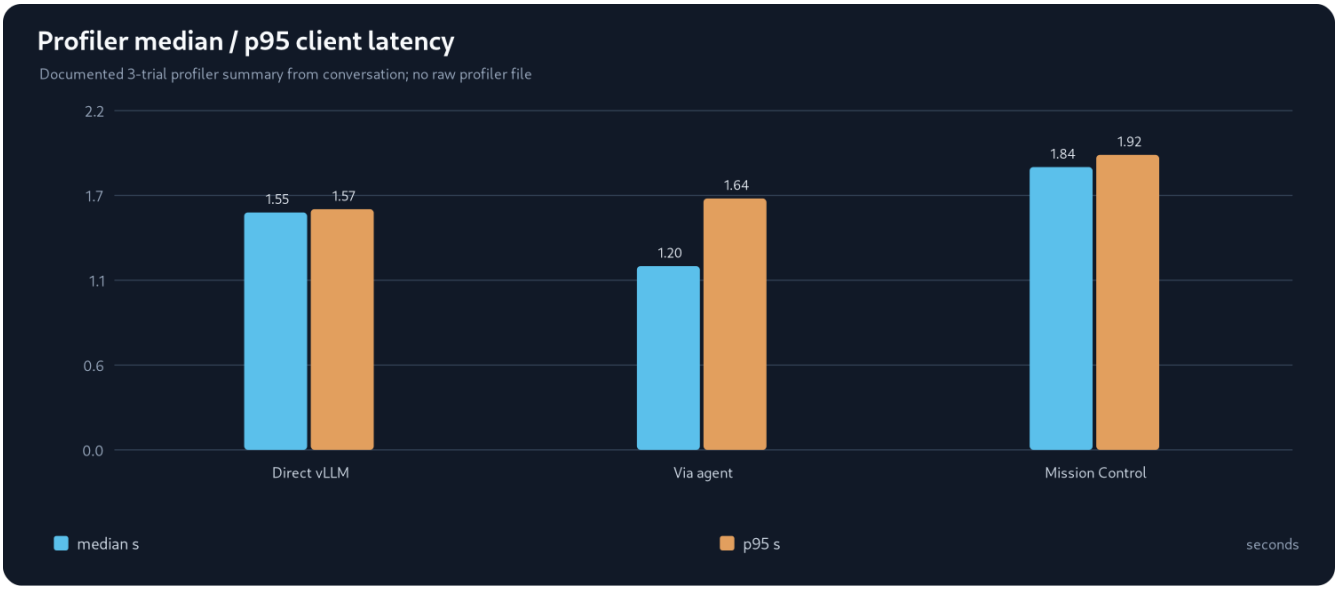
16 jobs show approximately eight pairs of job starts. Blue denotes inference; amber denotes additional post-start lifecycle.

13 / Dispatch wave intervals



Waves begin roughly every 1.4-2.2 seconds in the recorded 16-job trace.

14 / Profiler: direct, agent and controller



Profiler client latency includes Windows/SSH/network and scheduling overhead.

15 / Reproducibility and evidence index

The companion **Excel workbook** contains these structured sheets: START HERE, Real Run Index, Optimization, 16 Job Trace, Latency Profiler, Metric Dictionary, and Chart Data. Formula-based optimization deltas are editable.

The **offline Atlas generator** discovers local folders at results/mi300x, results/scheduler-lab, results/reliability, results/hardening and results/monitoring, and generates SVG figures, an interactive HTML gallery, ten indexed CSV tables, and a machine-readable manifest.

Data boundaries

• The multi-policy Scheduler Lab is CPU-only synthetic discrete-event simulation, not a physical MI300X comparison.

• The four virtual MI300X workers in the dashboard are modeled workers, not hardware partitions of one device.

• The physical MI300X deployment exposed one serving endpoint and therefore did not measure cross-GPU placement quality.

• Controller restart and worker drain results demonstrate tested simulation scenarios; they do not establish exactly-once real GPU execution.

• Raw per-run job IDs are redacted in the generated public index; no prompts, generated text, credentials or secret configuration files are copied.

Required additions for a final publication

Rerun the offline generator on the complete local archive to include all timestamped runs. Compare repeated identical runs and report trial-to-trial variation. Do not infer peak HBM or utilization from two to four telemetry samples. Validate all legends and throughput denominators before citing percentages.

Data cutoff: benchmark evidence shared through 27 October 2026. Prepared as an evidence-based preview.